

# Supplementary Material: Environment Sensitivity and Reproducibility of Post-BA SfM Benchmarks

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## A. What this document supports

This supplementary substantiates four specific statements in the main paper; a reviewer following any of its pointers can go directly to the listed section.

- Protocol, “Track provenance”: “ $\sim 0.209^\circ$  is the reproducible level [for the released RESfM checkpoint] ... unchanged ( $0.217^\circ$ ) under the authors’ exact pinned software stack” — Secs. B–D.
- Protocol: “a stack-replication study (supplementary) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism” — Sec. E.
- Table 2, † footnote: “TTT is run-to-run nondeterministic” — Sec. E.
- Reading any post-BA mean in this literature: differences  $\lesssim 0.05^\circ$  are within environment noise — Sec. F.

In addition, Sec. G tabulates the per-scene results behind the main paper’s dataset-level numbers, Sec. H gives the complete training and evaluation configuration, and Sec. I provides the per-scene label-degradation breakdown behind the label-ceiling analysis.

## B. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation  $1.89^\circ$  vs.  $1.29^\circ$ ). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry an irreducible environment-noise floor of  $\mathcal{O}(0.05)$ , produced by chaotic basin selection in robust bundle adjustment.

Table 1. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number.

| Condition                           | Rot ( $^\circ$ ) | Trans |
|-------------------------------------|------------------|-------|
| Published (paper Table 1)           | 1.29             | 0.169 |
| Our env (pip/OpenBLAS), 5-seed mean | 1.82             | 0.203 |
| Our env, seed 20                    | 1.89             | 0.209 |
| Authors’ exact stack (conda/MKL)    | 1.92             | 0.217 |

## C. Eliminating the controllable factors

**Data.** Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

**Randomness.** Five evaluation seeds vary the pre-BA network output by  $\pm 0.004$  and the post-BA mean by  $\pm 0.01$  — negligible.

**Our re-implementation.** Running the authors’ own evaluation code in our environment also yields  $\approx 0.209$ .

**Network forward pass.** The pre-BA output ( $11.88^\circ/1.730$ , identical across seeds) is deterministic; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

**Software stack.** We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives  $1.92^\circ/0.217$  — statistically indistinguishable from our environment’s  $1.89^\circ/0.209$ , and not the published  $1.29^\circ/0.169$ .

## D. Mechanism: chaotic basin selection in robust BA

Initialization perturbation is at the  $10^{-10}$  level. We isolated the BLAS layer by running the identical DLT triangulation (the BA initialization) on identical inputs under OpenBLAS and MKL, single-threaded: over 11,131 triangulated points the maximum relative difference is  $4.8 \times 10^{-10}$  (mean  $5 \times 10^{-14}$ ; NaN patterns identical).

Yet outcomes move by  $\mathcal{O}(0.1)$ . This sub-nanoscale perturbation changes the per-scene post-BA translation by up to 0.167, with 16/36 scenes moving by more than 0.01. Robust BA on ill-conditioned scenes amplifies last-bit differences into different convergence basins.

The published gap has the same signature. Comparing our released-checkpoint evaluation to the paper’s Table 1 per scene: 22/36 scenes agree closely (many exactly), 13 scenes are better in our runs, and the entire mean gap is carried by four basin flips (scenes 0046, 0148, 0012, 5016; e.g. 0046:  $0.043 \rightarrow 0.892$ ). Bidirectional, scene-concentrated flips are the signature of basin selection — not of a weaker checkpoint, which would degrade scenes uniformly.

## E. Stack replication and TTT run-to-run variance

Beyond the released checkpoint, we re-evaluated both from-scratch main-comparison checkpoints under the authors’ pinned stack (MKL numpy, pycolmap 3.10.0, pyceres 2.3), changing nothing else, and additionally repeated the TTT evaluation within our own stack. The supervised baseline is essentially unchanged across stacks—MegaDepth  $0.344 \rightarrow 0.350$  mean,  $0.070 \rightarrow 0.088$  median, only 7/36 scenes moving more than  $0.05^\circ$  (max  $\pm 0.37^\circ$ , bidirectional)—matching the released-checkpoint result above. The SS-RESfM weight+TTT arm, by contrast, varies regardless of stack: three evaluations of the same checkpoint (ours, a same-stack repeat, the authors’ stack) give MegaDepth means 0.316, 0.401, 0.453 ( $\sim 18$ – $20$ /36 scenes shifting per repeat, up to  $\pm 2.9^\circ$ , largely the same scenes, medians 0.12–0.19). The attribution is therefore not the software stack but test-time fine-tuning itself: it runs an optimization at evaluation time, so the chaotic basin selection documented above re-enters the result on every run. Consequently the paper reports all results from the single declared pipeline, treats TTT means as carrying  $\sim \pm 0.07^\circ$  run-to-run variance, and rests its environment- and run-independent claims on the deterministic no-TTT cells (which reproduce exactly on repeat).

Table 2. Per-scene 1DSfM results. Post-BA mean translation error (degrees) per scene, seed 20, for the three SS-RESfM arms and supervised RESfM (all from-scratch, RESfM-aligned schedule). Best per scene in bold.

| Scene               | madweight | weight | weight+TTT | RESfM |
|---------------------|-----------|--------|------------|-------|
| Alamo               | 1.77      | 2.18   | 1.67       | 1.73  |
| Ellis_Island        | 17.59     | 15.90  | 17.11      | 16.02 |
| Madrid_Metropolis   | 9.63      | 4.37   | 3.52       | 3.38  |
| Montreal_Notre_Dame | 0.94      | 11.35  | 1.24       | 0.93  |
| Notre_Dame          | 0.68      | 0.34   | 0.33       | 0.49  |
| NYC_Library         | 2.94      | 1.19   | 3.39       | 6.28  |
| Piazza_del_Popolo   | 0.94      | 10.66  | 17.05      | 2.49  |
| Tower_of_London     | 76.34     | 28.29  | 72.61      | 67.95 |
| Vienna_Cathedral    | 7.30      | 22.71  | 33.57      | 2.29  |
| Yorkminster         | 5.79      | 30.87  | 10.79      | 8.77  |

## F. Implications for reading the paper’s numbers

(1) The published number and ours are draws from a distribution over environments; no software matching can force a particular draw. (2) Differences of  $\lesssim 0.05$  in post-BA translation means on these benchmarks are within environment noise and should not be read as method differences — by any paper. (3) Consequently, meaningful comparisons must be environment-controlled: every number in the main paper (all arms, RESfM-from-scratch, and the released checkpoint) is produced in a single environment, and the out-of-distribution margins ( $2\times$  and larger) sit far above this noise floor.

## G. Per-scene results

Tables 2–4 report the per-scene post-BA mean translation errors underlying the main paper’s dataset-level means and medians (seed 20, the declared pipeline): the ten 1DSfM scenes and the 36 MegaDepth test scenes, for the three self-supervised arms (madweight, weight, weight+TTT) and the supervised RESfM baseline trained from scratch under the identical recipe. The per-scene view makes two aggregate statements concrete: on 1DSfM the self-supervised wins are broad-based rather than carried by a single scene (with Ellis\_Island and Tower\_of\_London hard for every arm), while on MegaDepth the four arms are close on most scenes and the dataset-level gap is dominated by a handful of high-error scenes.

## H. Implementation details

Table 5 lists the full training and evaluation configuration. Both methods share every setting except the base learning rate ( $10^{-4}$  self-supervised,  $10^{-3}$  supervised, each selected on its own training objective) and the supervision source (COLMAP-derived labels vs. the adaptive percentile pseudo-labels).

Table 3. Per-scene MegaDepth results (1/2). Post-BA mean translation error (degrees) per scene, seed 20. Best per scene in bold. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. E).

| Scene | madweight | weight | weight+TTT | RESfM |
|-------|-----------|--------|------------|-------|
| 0238  | 1.41      | 0.62   | 0.63       | 0.12  |
| 0060  | 0.03      | 0.02   | 0.37       | 0.01  |
| 0197  | 0.89      | 0.38   | 1.11       | 0.11  |
| 0094  | 0.45      | 0.16   | 0.17       | 0.88  |
| 0265  | 3.26      | 2.34   | 0.19       | 3.09  |
| 0083  | 0.23      | 1.56   | 0.20       | 0.10  |
| 0076  | 0.38      | 0.23   | 0.22       | 0.02  |
| 0185  | 0.01      | 0.04   | 0.09       | 0.01  |
| 0048  | 0.14      | 0.02   | 0.03       | 0.11  |
| 0024  | 0.18      | 0.17   | 1.13       | 1.09  |
| 0223  | 0.24      | 0.23   | 0.30       | 0.08  |
| 5016  | 0.04      | 0.02   | 0.02       | 0.01  |
| 0046  | 0.22      | 0.30   | 0.25       | 0.93  |
| 0099  | 0.15      | 3.52   | 0.13       | 0.39  |
| 1001  | 1.22      | 0.73   | 0.49       | 0.50  |
| 0231  | 0.15      | 0.07   | 0.24       | 0.02  |
| 0411  | 0.03      | 0.32   | 0.46       | 0.02  |
| 0377  | 0.05      | 0.26   | 0.03       | 0.06  |

Table 4. Per-scene MegaDepth results (2/2). Continuation of Table 3.

| Scene | madweight | weight | weight+TTT | RESfM |
|-------|-----------|--------|------------|-------|
| 0102  | 0.02      | 0.01   | 0.01       | 0.02  |
| 0147  | 2.99      | 1.21   | 1.20       | 3.22  |
| 0148  | 0.33      | 0.11   | 0.04       | 0.04  |
| 0446  | 0.03      | 0.19   | 0.19       | 0.05  |
| 0022  | 0.02      | 0.03   | 0.04       | 0.02  |
| 0327  | 0.09      | 0.02   | 0.02       | 0.01  |
| 0015  | 0.30      | 0.56   | 0.65       | 0.30  |
| 0455  | 0.01      | 0.01   | 1.60       | 0.09  |
| 0496  | 0.04      | 0.04   | 0.09       | 0.07  |
| 1589  | 0.06      | 0.05   | 0.30       | 0.01  |
| 0012  | 0.02      | 0.70   | 0.22       | 0.34  |
| 0104  | 0.03      | 0.18   | 0.07       | 0.02  |
| 0019  | 0.17      | 0.28   | 0.13       | 0.01  |
| 0063  | 0.08      | 0.07   | 0.06       | 0.07  |
| 0130  | 1.60      | 0.16   | 0.01       | 0.07  |
| 0080  | 0.05      | 0.20   | 0.07       | 0.10  |
| 0240  | 0.25      | 0.34   | 0.46       | 0.35  |
| 0007  | 0.01      | 0.09   | 0.14       | 0.04  |

Table 5. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

| Setting                                | Value                                                                    |
|----------------------------------------|--------------------------------------------------------------------------|
| Training (both methods)                |                                                                          |
| Epochs                                 | 20,000 (one epoch = one pass over all scenes)                            |
| Batch                                  | 4 scenes                                                                 |
| LR schedule                            | milestones 10k, 11k, ..., 20k, $\gamma = 0.9$                            |
| Base LR                                | $10^{-4}$ (SS-RESfM) / $10^{-3}$ (RESfM)                                 |
| Loss weights                           | $\alpha_{\text{reproj}} = 1, \beta_{\text{cls}} = 1$                     |
| Infinity-point margin                  | $10^{-4}$                                                                |
| Checkpoint selection                   | best validation error, evaluated every 500 epochs                        |
| Seed                                   | 20 (21–22 / 21–24 for the seed studies)                                  |
| Adaptive pseudo-labels (SS-RESfM only) |                                                                          |
| Inlier / outlier percentile            | 20 / 80 (of per-observation reproj. error)                               |
| Threshold separation                   | scale-aware floor, $0.25 \times$ median error                            |
| Warm-up                                | 10 epochs before pseudo-labeling starts                                  |
| Evaluation                             |                                                                          |
| Removal threshold $\tau$               | 0.6 (0.8 for low-contamination single-camera sets, following RESfM)      |
| Test-time fine-tuning                  | 1,001 iterations, LR $5 \times 10^{-3}$ , evaluated every 250 iterations |
| Bundle adjustment                      | pyceres/pycolmap robust BA, as RESfM                                     |

Table 6. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Recall collapses above  $\sim 15\%$  contamination.

| Dataset    | Scene             | Contam. (%) | Coverage | Prec. | Recall |
|------------|-------------------|-------------|----------|-------|--------|
| BlendedMVS | 58c4bb4f4a...     | 0.5         | 1.00     | 0.86  | 0.94   |
| Strecha    | Herz-Jesu-P8      | 0.7         | 1.00     | 0.51  | 0.66   |
| Strecha    | fountain-P11      | 0.7         | 1.00     | 0.69  | 0.76   |
| Strecha    | Herz-Jesu-P25     | 2.6         | 0.99     | 0.72  | 0.82   |
| Strecha    | entry-P10         | 2.7         | 0.99     | 0.88  | 0.88   |
| BlendedMVS | 5be883a4f9...     | 4.0         | 0.98     | 0.90  | 0.84   |
| BlendedMVS | 5b6e716d67...     | 6.1         | 0.92     | 0.89  | 0.92   |
| BlendedMVS | 5bfd0f32ec...     | 7.1         | 0.97     | 0.97  | 0.97   |
| BlendedMVS | 5b950c7160...     | 9.2         | 0.96     | 0.92  | 0.94   |
| MegaDepth  | 0007              | 11.2        | 0.73     | 0.84  | 0.47   |
| 1DSfM      | Alamo             | 26.6        | 0.76     | 0.81  | 0.25   |
| 1DSfM      | Madrid_Metropolis | 28.8        | 0.67     | 0.85  | 0.28   |
| MegaDepth  | 0041              | 31.4        | 0.49     | 0.80  | 0.42   |
| MegaDepth  | 0060              | 39.0        | 0.42     | 0.53  | 0.45   |
| 1DSfM      | Piazza_del_Popolo | 41.9        | 0.76     | 0.90  | 0.30   |
| MegaDepth  | 0047              | 54.9        | 0.37     | 0.83  | 0.50   |
| 1DSfM      | Ellis_Island      | 57.9        | 0.79     | 0.89  | 0.37   |
| 1DSfM-hard | Gendarmenmarkt    | 59.8        | 0.50     | 0.89  | 0.29   |

## I. Per-scene label degradation

Table 6 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below  $\sim 10\%$  contamination the labels are near-complete and high-recall, while every scene above  $\sim 15\%$  contamination shows recall collapsing toward  $\sim 0.3$  with coverage dropping toward  $\sim 0.55$  — the supervision signal degrades exactly where outlier handling matters most.